MIL-M-80016B
20 July 1972
SUPERSEDING
MIL-M-80016A
1 April 1967
MIL-M-13819A(ORD)
2 June 1959

MILITARY SPECIFICATION

MILLING MACHINE, VERTICAL, SWIVEL HEAD, KNEE AND COLUMN, RAM AND TURRET TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers vertical, swivel head, plain table, ram and turret type, electric motor driven, floor mounted milling machines, with equipment and accessories as described herein.

* 1.2 Classification. The milling machines covered by this specification shall be of the following sizes and classes as specified (see 6.2).

Size 1 - 1 horsepower spindle drive
- Class 1 - 8 inches minimum table width
- Class 2 - 9 inches minimum table width

Size 2 - 1-1/2 horsepower spindle drive
- Class 1 - 9 inches minimum table width
- Class 2 - 10 inches minimum table width

Size 3 - 2 horsepower spindle drive
- Class 1 - 10 inches minimum table width
- Class 2 - 12 inches minimum table width

Size 4 - 3 horsepower spindle drive
- Class 1 - 12 inches minimum table width

Size 5 - 4 horsepower spindle drive
- Class 1 - 11 inches minimum table width

2. APPLICABLE DOCUMENTS

* 2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 3417

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SPECIFICATIONS

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 Westpark Dr., McLean, Virginia 22101.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360.

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Mass. Ave., N. W., Washington, D. C. 20005.)

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AMERICAN NATIONAL STANDARDS

ANS B5.1	T-Slots - Their Bolts, Nuts, Tongues and Cutters
ANS B5.10	Machine Tapers, Self-Holding and Steep Series
ANS B5.18	Spindle Noses and Arbors for Milling Machines

(Application for copies should be addressed to the American National Standards Institute, Department 969, 1430 Broadway, New York, N. Y. 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First Article approval. If First Article approval is required (see 6.2 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements hereinbefore or in the initial stage of, production under a contract.

* 3.2 Design. The machine shall be new and one of the manufacturer's current models. The design shall provide convenient and safe operation. All parts of the machine shall be new, designed and constructed to requirements listed herein. All parts subject to wear, breakage, or distortion shall be easily accessible for adjustment, replacement and repair.

* 3.2.1 Maintainability. Item configuration and adjustment features shall be designed to reduce maintenance operations to the simplest procedures compatible with a minimum of maintenance skills and common maintenance equipment. Instructions for disassembly, repairs, adjustments, and reassembly will be concise and clearly defined.

* 3.2.2 Safety measures. Guarding requirements and safety devices for the machine, components thereof, and machine functions that present personnel hazards shall be in accordance with the applicable portions of the latest Occupational Safety and Health Act (currently 1970). Additional requirements for safety and health shall be as specified (see 6.2 and 6.5).

* 3.2.3 Lubrication. A lubrication system equipped with a screen or filter shall be provided to insure a supply of clean oil to all mating gears,

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sliding parts, and bearings except sealed-for-life type. Forced feed metered lubrication shall be provided for ways and remote areas, with controls mounted on the machine convenient to the operator. A one-shot lubrication system shall be considered satisfactory for this purpose. Fully automatic lubrication of gears and fully automatic or forced feed metered lubrication of feed screw shall be provided. Reservoirs shall have oil level sight gages. All oil holes, grease fittings, and filler caps shall be easily accessible.

* 3.2.4 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

* 3.3 Construction. The machine shall be so constructed that when installed and connected to power, it will be ready for operation. The machine shall be constructed of parts which are without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

* 3.3.1 Casting and forgings. All castings and forgings shall be free from defects, scale and mismatching. No process such as welding, peening, plugging or filling with solder or metallic paste shall be used for reclaiming any defective part for use in the machine.

* 3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where required in the original design. None of these operations shall be employed as a repair measure for any defective part.

* 3.3.3 Fastening devices. All screws, pins, bolts, and similar internal and external parts shall be installed with means for preventing change of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, sprues, fins, flash, scale, flux and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

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* 3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

* 3.3.7 Feed indicators. All feeds of the machine shall have suitable feed indicators which utilize the inch system, the smallest digit or graduation representing 0.001 inch or less of stock removal. Indicator graduations shall be easily read.

* 3.3.8 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language. Letters shall be engraved, etched, embossed or stamped in bold-face on a contrasting background.

* 3.3.9 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

* 3.4 Components. The milling machines shall consist essentially of a ram swivel head, ram turret, table, saddle, knee, column, and base and such other features as are necessary to meet the performance requirements specified herein.

3.4.1 Ram. The ram shall be designed for mounting the plain or universal swivel head as applicable. The ram shall have matching slides machined and fitted to the turret ways. Means shall be provided for clamping and locking in any desired position within the rated ram travel. Ram positioning shall be by manual mechanical means.

3.4.2 Ram turret. The ram turret shall be machined and fitted to the top of the column. The turret shall have means for clamping in any position within a 180-degree horizontal plane. The turret shall be provided with suitable markings to indicate zero swivel. The top of the turret shall be designed with dovetail or square lock ways with matching slides on the ram.

* 3.4.3 Table. The table shall be of one-piece construction. The bottom of the table shall be machined and fitted to ways of the saddle. It shall have a locking arrangement for clamping to the saddle. The table shall be of the trough type with coolant channels and a drain hole fitted with a return line to the reservoir. Table surface shall be machine finished and provided with not less than three equally spaced T-slots. The front

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side of the table shall be a finished surface with a T-slot running the full length of the table for mounting adjustable stops. Two adjustable table stops shall be furnished. Manual longitudinal table feed shall be by means of a feed screw operated by a handwheel or crank and shall be fitted with a dial graduated in thousandths of an inch and marked in ten thousandths increments. When specified (see 6.2), power feed to the table shall be furnished. Means shall be provided to prevent overtravel of the table feed mechanism.

* 3.4.4 Saddle. The saddle shall have strength and proportions suitable for supporting the table. The saddle shall be equipped with tapered adjustable gibs and guideway wipers or other suitable means for preventing dust and chips from entering ways. Bearing surfaces and ways shall be machined and scraped to fit mating parts. The saddle shall have means for lubrication of all knee, saddle and table bearing surfaces and moving mechanisms in the saddle. Manual saddle cross feed shall be provided. The feed screw shall be heat-treated steel with acme or square threads and shall be fitted with a dial graduated in thousandths or less to indicate saddle cross feed. When specified (see 6.2), power feed to the saddle shall be furnished for Size 2, Class 2; Size 3, Class 2; and all Sizes 4 and 5 machines. The cross feed handwheel shall be automatically disengaged when power rapid traverse is engaged.

3.4.5 Knee. The knee shall be machined and fitted to the column ways with adjustable gibs. The knee shall support the saddle and table and shall have vertical movement by means of an elevating screw operated by a crank or handwheel mechanism. When specified (see 6.2), power elevation of the knee shall be supplied to Size 2, Class 2, and all Sizes 3, 4 and 5 machines.

3.4.6 Column and base. The column and base shall be of such design and construction as to minimize vibration and assure the accuracies and performance specified herein. The front of the column shall have machined and fitted ways for the knee. The top of the column shall be machined and fitted to the ram turret. The periphery of the column top shall be marked to duplicate the zero setting marks on the ram turret. The base shall have provisions for securing the machine to the floor or foundation.

* 3.4.7 Swivel head. Each machine shall have a swivel head with one or more graduated scales or dials to indicate the angular position of the head in the plane, or planes, in which it moves. Unless otherwise specified (see 6.2), the head shall be provided with a mechanism for fast movement of quill, a handwheel or knob for slow quill movement, a micrometer depth stop, a locking device and manual spindle feed. When specified (see 6.2), power feed to the quill shall be furnished for all sizes except Size 4.

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* 3.4.8 Spindle. The spindle shall be of heat-treated alloy steel journalled on precision type ball or roller bearings. Means shall be provided for axial adjustment of the bearings. The spindle nose shall be precision ground. It shall have a standard taper conforming to ANS B5.10 or B5.18, or shall be equipped with a collet type holder with draw-bar or other positive means of closing the collet for firmly holding the tool. The spindle taper of the machine may be any of the tapers listed for its class in Table I, or when so indicated in Table I, may be of the collet type with indicated minimum collet capacity. Spindle taper of Size 1, Class 1 machines shall be R-8, No. 30 AN Standard, or No. 9 Brown and Sharpe, as specified (see 6.2).

* 3.5 Sizes and capacities. Sizes and capacities of the milling machines covered by this specification shall meet the requirements of Table I.

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TABLE I

SIZES AND CAPACITIES

CHARACTERISTICS	Size 1		Size 2		Size 3		Size 4	Size 5
	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Class 1	Class 1
Table working surface								
Length by width, inches								
Standard size	32 x 8	36 x 9	36 x 9	40 x 10	40 x 10	54 x 12	40 x 12	58 x 11
Optional table size No. 1	36 x 8	42 x 9	42 x 9	48 x 10	48 x 10	60 x 12	50 x 12	-
Optional table size No. 2	-	46 x 9	46 x 9	-	-	-	-	-
T-Slots, number and size, inches	3, 11/16	3, 5/8	3, 5/8	3, 11/16	3, 11/16	3, 11/16	3, 11/16	3, 5/8
Table feed, inches								
Longitudinal								
with standard table	20	20	20	20	20	28	20	30
with optional table size No. 1	25	26	26	28	28	34	28	-
with optional table size No. 2	-	32	32	-	-	-	-	-
Cross								
Standard	8	9	9	10	10	10	10	15
Optional	13	12	12	-	-	-	-	-
Vertical	15	16	16	16	15	18	17	16
Spindle speeds, number of	8	8	8	Infinite	8	Infinite	12	Infinite
Optional number of speeds	-	Infinite	Infinite	-	Infinite	-	Infinite	-
Spindle speed range	80 to 2700	80 to 2700	160 to 2700	120 to 3300	300 to 4300	85 to 3800	80 to 2100	50 to 3500
Quill feeds								
Travel, inches	4	5	5	4.5	4	5	-	5
Power feeds, number of, when specified under 3.4.7	Infinite	3	3	3	Infinite	3	-	Infinite
Quill feed range								
Inches per revolution or inches per minute	0.5 to 2.0 IPM	.0015 to .006 IPR	.0015 to .006 IPR	.0015 to .006 IPR	0.25 to 3.0 IPM	.0015,.003, .006 IPR	-	1/16 to 2-1/2 IPM
ANS spindle taper	(See 3.4.8)	R-8, 9B&S	R-8, 9B&S	30NS,40NS,R-8	30NS, 40NS	40NS	40NS	40NS
Collet capacity, when furnished, inches	-	3/4	3/4	-	-	-	1	-
Head swivel along longitudinal axis, degrees	*90°-0-90°	90°-0-90°	90°-0-90°	90°-0-90°	90°-0-90°	90°-0-90°	90°-0-90°	90°-0-90°
Head swivel across longitudinal axis, degrees	45°-0-45°	45°-0-45°	45°-0-45°	-	-	-	-	30°-0-30°
Spindle motor horsepower	1	1	1-1/2	1-1/2	2	2	3	4
Longitudinal power feed rates, number of, when specified under 3.4.3	Infinite	8	8	9	Infinite	16	Infinite	Infinite
Longitudinal, IPM	1 to 30	1 to 9	1 to 9	3/4 to 15	3/4 to 20	1/4 to 30	3/4 to 20	3/4 to 26
Cross, IPM, when specified under 3.4.4	1 to 30	None	None	3/4 to 15	None	1/4 to 30	3/4 to 20	3/4 to 26
Vertical, IPM, when specified under 3.4.5	1 to 30	None	None	1/2 to 10	1/2 to 10	1/8 to 15	1/2 to 10	3/8 to 13
Net weight, pounds	1250	1800	1900	2450	2800	3700	2950	5200

* 30-0-90 with powerfeed quill.

Requirements are not less than those shown. Where a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

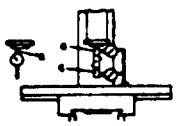
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TABLE II

MILLING MACHINE, RAM TYPE, SWIVEL HEAD

ALIGNMENT TEST

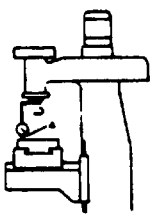
Fig.1 - Spindle face and radial runout



Allowable deviation, not more than:

A. 0.005" (TIR)
B. 1" from spindle nose 0.0004" (TIR)
C. 6" from spindle nose 0.0005" (TIR)

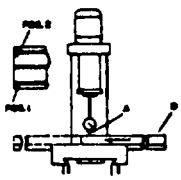
Fig.2 - Spindle square with table surface



Allowable deviation not more than:

0.001" per foot
Table low in rear

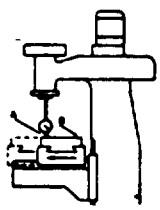
Fig.3 - Longitudinal rise and fall of table



Allowable deviation, for both positions, not more than:

0.001" per 12"

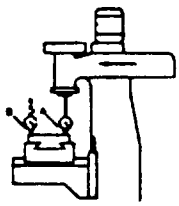
Fig.4 - Transverse rise and fall of table



Allowable deviation not more than:

0.001" per 12"
Table low in rear

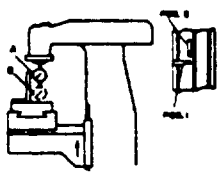
Fig.5 - Ram travel parallel with table surface



Allowable deviation not more than:

0.002" in 7"

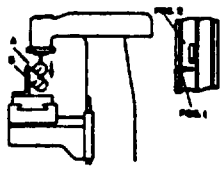
Fig.6 - Knee travel parallel to spindle



Allowable deviation for traverse and longitudinal positions, not more than:

0.001" per foot

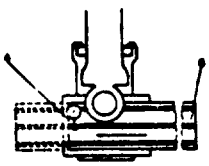
Fig.7 - Quill movement square with table



Allowable deviation for both positions, not more than:

0.001" per 4"

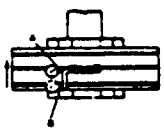
Fig.8 - Center T-slot parallel to table travel



Allowable deviation, not more than:

0.001" per foot

Fig.9 - Center T-slot square with cross travel



Allowable deviation, not more than:

0.001" per foot

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* 3.6 Performance. Each machine shall be capable of satisfactory performance when tested in accordance with 4.5.1 through 4.5.4.3.

3.7 Alignment tolerances. The alignment tolerances of the machine shall comply with the requirements of Table II.

* 3.8 Electrical equipment. Unless otherwise specified (see 6.2), the electrical system shall conform to JIC Electrical Standards for General Purpose Machine Tools. Unless otherwise specified (see 6.2), the machine shall operate from 230/460 volt, 3 phase, 60 Hz power supply, initially wired for operation on the voltage specified (see 6.2). An identified terminal shall be provided for grounding the machine when installed. The terminal shall be suitable for connection of the size grounding conductor specified in Table 15-1 of JIC Standards.

* 3.8.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall meet the requirements of Table I. When specified (see 6.2), all motors shall be squirrel cage induction type. Unless otherwise specified (see 6.2), motors shall meet the requirements of a dripproof enclosure.

* 3.9 Standard equipment. The following equipment shall be furnished with each machine: one complete set of wrenches normally provided by the manufacturer.

TABLE I

* 3.10 Optional equipment. When specified (see 6.2), the following optional equipment and accessories shall be furnished.

3.10.1 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and piping. The reservoir shall incorporate a strainer or settling chambers to remove cutting chips from the coolant. The reservoir shall be equipped with a threaded drain plug. The coolant system shall include oil resistant hoses, tubing, or piping to deliver the coolant. The coolant delivery nozzle shall be fitted with a valve to permit the operator to direct and control the amount of flow.

3.10.2 Measuring equipment. The milling machine shall be equipped with end measuring rod and indicators, or optical vernier type measuring equipment as specified.

* 3.10.3 Raising block. The milling machine shall be equipped with a raising block or blocks between the top of the column and the ram arm turret at least of the height specified.

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* 3.10.4 Slotter head. The milling machine shall be equipped with a slotter head on the ram opposite the milling head.

* 3.10.5 Angle milling attachment. The milling machine shall be equipped with an angle milling attachment as specified.

3.11 Additional optional equipment. Additional optional equipment shall be furnished as specified (see 6.2).

* 3.12 Electromagnetic interference control. Unless otherwise specified (see 6.2), equipment procured under this specification shall comply with the requirements of MIL-STD-461 for Class IIB equipment.

* 3.13 Fungus control. When required (see 6.2), fungus proofing shall be as specified.

* 3.14 Lubrication chart or plate. Unless otherwise specified (see 6.2), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

Federal or Military Specification Number

* 3.15 Nameplate. Unless otherwise specified (see 6.2), a corrosion-resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity.

Nomenclature

Manufacturer's name

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Manufacturer's model designation

Machine size

Manufacturer's serial number

Power input (volts, total amps, phase, frequency)

Contract Number or Order Number

Federal Stock Number or Plant Equipment Code

Date of manufacture

U S

* 3.16 Identification plate. When specified (see 6.2), a separate identification plate of corrosion-resistant metal permanently and legibly marked in letters 3/16 inch high with the caption "Departmental or Activity Identification" shall be securely attached to each machine. The plate shall have blank space for the government to stamp 25 characters of the same size as the caption.

* 3.17 Technical data. Data shall be furnished as specified (see 6.2.2).

* 3.18 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the First Article.

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* 4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.4.3 and the inspection in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.18.

4.5 Tests.

* 4.5.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be determined during the trial period.

* 4.5.2 Alignment tolerance. The machine shall be tested for compliance with 3.7.

* 4.5.3 Metal removal and tool chatter test. The machine shall be used to mill a workpiece of SAE 1045 steel. The workpiece shall not be less than twelve inches long by three inches wide by two inches thick. One of the surfaces having the largest area shall be ground flat and mounted in contact with the table. The workpiece may be drilled to facilitate clamping. The machine shall make a roughing cut to mill metal off the entire horizontal surface of the workpiece in one pass at sufficient depth and feed rate to load the spindle drive motor to not less than its maximum continuous rated electrical input. The spindle speed shall not exceed 10 percent of the highest speed in Table I for the machine required. Except as limited in this paragraph, the size of the workpiece, the spindle speed, type of cutting tool, and table feed rate shall be at the discretion of the supplier. The resultant metal removal rate shall be not less than 0.65 cubic inches per minute per rated horsepower of the spindle drive motor. The surface of the workpiece after the cut shall show no evidence of tool chatter.

* 4.5.4 Machining accuracy tests. Tests shall be made to determine machining accuracy as follows. Using the same workpiece as 4.5.3, a roughing cut shall be made on one vertical face of the workpiece. Finishing cuts shall then be made on the horizontal and vertical faces which were previously rough cut.

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4.5.4.1 Squareness. The two milled sides shall be checked for squareness. The adjacent milled sides shall be square with each other within .001 inch per four inches.

4.5.4.2 Flatness. The two milled sides shall be tested for flatness. Each side shall be flat within .001 inch, concave only.

4.5.4.3 Parallelism. The horizontal milled surface shall be checked for parallelism with the ground surface. Surfaces shall be parallel within 0.001 inch per foot. Parallelism may be determined by relation to surface of worktable with test piece clamped in position, or by thickness measurement after removal from the worktable.

* 4.5.5 Electromagnetic interference control tests. Unless otherwise specified (see 6.2), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.12 in accordance with Table II of MIL-STD-461.

4.6 Inspection of preparation for delivery. Preservation-packaging, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2).

* 5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B or C as specified (see 6.2).

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Additional marking shall be as specified (see 6.2).

6. NOTES

* 6.1 Intended use. The milling machines covered by this specification are to be used in the shops and depots of activities in the Department of Defense.

* 6.2 Ordering data. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

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* 6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Size and class of machine required (see 1.2).

c. First Article approval, if required (see 3.1).

d. Additional safety and health requirements (see 3.2.2 and 6.5).

e. Painting, if different (see 3.3.5).

f. Power feed to table, if required (see 3.4.3).

g. Power feed to saddle, if required (see 3.4.4).

h. Power elevation of knee, if required (see 3.4.5).

i. Swivel head requirement, if different (see 3.4.7).

j. Power feed to quill, if required (see 3.4.7).

k. Type spindle nose required for Size 1 Class 1 machines (see 3.4.8).

l. Options, if required (see Table I).

m. Electrical equipment, if different (see 3.8).

n. Voltages of machine power supply, if different (see 3.8).

o. Voltage for initial wiring of machine (see 3.8).

p. Squirrel cage induction type motors, if required (see 3.8.1 and 6.6).

q. Type enclosure for motors, if different (see 3.8.1).

r. Optional equipment required (see 3.10).

s. Additional optional accessories and equipment, if required (see 3.11).

t. Electromagnetic interference control, if different (see 3.12 and 6.4).

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u. Type fungus proofing, if required (see 3.13).

v. Lubrication chart or plate, if different (see 3.14).

w. Nameplate, if different (see 3.15).

x. Identification plate, if required (see 3.16).

y. First Article inspection, if different (see 4.2).

z. Quality conformance inspection, if different (see 4.3).

aa. Electromagnetic interference control test requirements, if different (see 4.5.5 and 6.4).

bb. Required level of preservation-packaging (see 5.1).

cc. Required level of packing (see 5.2).

dd. Additional marking, if required (see 5.3).

* 6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.17).

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should include complete information concerning examination, testing, and approval.

* 6.4 Army and Air Force procurements. Equipment purchased for the Departments of the Army and the Air Force should have electromagnetic interference control tests in conformance with MIL-STD-461, Class IIB equipment, except that only Test Methods CE03 and RE02 should be required.

* 6.5 Safety and health environmental determinations. In order that equipment may be properly integrated into the user's operational environment in compliance with the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning guarding the point(s) of operation, and other safety requirements should be specified.

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* 6.6 Squirrel cage induction motors. Procuring activities should exercise due caution in requiring this motor type (see 3.8.1 and 6.2.1.q), particularly when variable speed motors are involved. The requirement may eliminate the commercial equipment of some, or all, potential suppliers.

* 6.7 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - WC
Navy - SH, AS, OS
Air Force - None

User Activities:
None

Preparing Activity:
DSA-IP

Project Number 3417-0039

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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